Multitemporal Symmetric Fusion Network for Hyperspectral Change Detection

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***Abstract*— Hyperspectral images (HSIs) offer detailed and abundant spectral–spatial information, holding great potential for ground object change detection (CD). Currently, numerous deep learning-based hyperspectral CD (HCD) models have been studied. However, these methods only consider single difference information or temporal information while neglecting the complementary advantage between difference and temporal information. To solve this issue, this work proposes a multitemporal symmetric fusion network (MTSFNet) for HCD, which involves three steps. First, the difference and temporal data are calculated by a subtraction operation and concatenation operation. Then, a dual-branch convolutional network is developed to capture the difference and temporal features. Finally, a symmetric feature fusion scheme is designed to integrate the extracted features followed by a fully connected layer to derive the detection results. Experiments conducted on multiple well-known datasets reveal that the proposed MTSFNet outperforms other advanced CD approaches in terms of qualitative and quantitative results.**

***Index Terms*— Deep learning, hyperspectral change detection (HCD), hyperspectral images (HSIs), information fusion, remote sensing.**

I. Introduction

REMOTE sensing change detection (CD) seeks to detect and locate different regions in multitemporal remote sensing images [1], which is a crucial step in several applications, such as land survey [2], [3], marine monitoring [4], [5], and military reconnaissance [6], [7]. Due to advancements in remote sensing imaging technology, there is an increasing ability to acquire a greater number of hyperspectral images (HSIs) [8]. Recently, hyperspectral remote sensing have garnered growing interest in detecting the changed regions, since HSI provides finer spectral–spatial information of land covers [6], [9]. Hyperspectral CD (HCD) techniques enable the detection of changes at a fine-grained level, facilitating the identification of subtle modifications that may not be easily discernible in traditional multispectral imagery [10].

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Over the past few decades, many traditional image processing techniques have been applied in CD [11], including band algebraic calculation, feature transformation, and index extraction. Classical CD algorithms primarily compare the difference of spectral reflectance in original images, including difference vegetation index (DVI) as well as normalized vegetation index (NDVI) [12]. Following this idea, numerous CD techniques have been created, including band difference method, ratio method, regression analysis, independent component analysis (ICA) [13], principal component analysis (PCA) [14], iteratively reweighted MAD (IRMAD) [15], change vector analysis (CVA) [16], and multivariate alteration detection (MAD) [17]. These techniques are primarily oriented toward remote sensing images with moderate and lower resolutions, often overlooking the spatial correlation between pixels during the CD process. Moreover, the detection performance of these techniques can be greatly affected by the predetermined thresholds used in the detection process.

Besides, many machine-learning methods have been applied for CD due to their ability to analyze complex data patterns, including support vector machines (SVMs) [18], random forests (RFs) [19], artificial neural networks (ANNs) [20], decision trees (DT) [21], and conditional random field (CRF) [22]. Furthermore, various improved techniques have been studied. For example, Lu et al. [23] first studied a guided filtering-based HCD method, where the guided filtering was used to preserve the edges of changed regions. Hou et al. used multiple morphological profiles to extract the differential features [24]. These methods can improve the detection effect. However, they seriously rely on the handcrafted feature extraction techniques.

The rapid development of deep learning has driven significant advancements in fields such as hyperspectral imaging [25] and anomaly detection [26]. Alongside the rapid evolution of aerospace observation technology, the emergence of remote sensing pretrained models like SpectralGPT enables the comprehensive extraction and analysis of hyperspectral data [27]. This significantly enhances the accuracy and generalization capabilities in image processing tasks. In the last few years, there has been a growing trend of deep-learning models for HCD [28], [29], which can be roughly grouped into two types. One type of CD methods are first to obtain stacked or difference image, and then, the stacked or difference image is given as input to a deep neural network to gain the detection result. For example, Ou et al. [30] presented an improved slow feature analysis technique to differentiate unchanged

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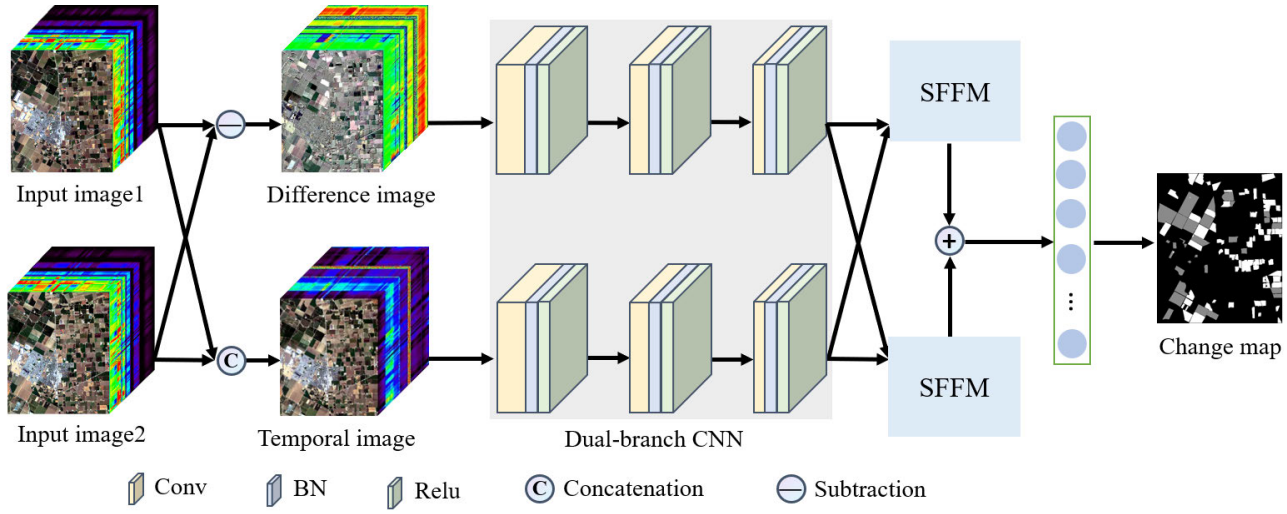


Fig. 1. Schematic of the proposed MTSFNet.

and changed areas, in which an feature fusion approach was exploited to enhance the discrimination. Li et al. [31] introduced a novel approach by combining two model-driven schemes, i.e., structural similarity and CVA, to derive reliable pseudolabels for learning a convolutional neural network (CNN). Zhang et al. [32] introduced a cascade spectral-guided transformer (CAST) architecture to achieve CD by fully exploiting the temporal and spectral–spatial information.

Another type of CD methods are primarily to generate the change feature on single source image, and these change features are then subjected to a difference operation, followed by a classifier to identify difference regions. For example, Zhan et al. [33] applied a Siamese architecture-based CNNs for HCD, which utilized tensor pairs extracted from HSIs to determine the similarity and identify changed areas. Lin and Lin [34] proposed CODE-HCD, a novel approach for HCD, which utilized the convex optimization and deep learning to address the challenges of relying on large training data and complex mathematical regularization terms. Zhao et al. [35] proposed a learning interpretable network, in which the CD model was converted into an interpretable neural network. Wang et al. [36] devised a joint spectral, spatial, and temporal long-range attention model for HCD, which effectively leveraged spectral, spatial, and temporal information to detect the changed areas.

Although these approaches can identify the changed regions, they still face several challenging issues: 1) these methods mentioned above only consider either change or temporal information while neglecting the complementary information between changed and temporal features, which cannot effectively utilize the temporal spatial–spectral information and 2) most of existing methods exploit simple subtraction operation to obtain the fused features, which cannot sufficiently improve the capability to discriminate changed and unchanged areas.

To solve these issues, we propose a multitemporal symmetric fusion network (MTSFNet) for HCD. First, the difference and temporal images are obtained with subtraction and concatenation operations. Then, a dual-branch CNNs is employed to obtain the difference and temporal features. Finally, the multitemporal symmetric feature fusion module is leveraged to amalgamate the extracted features, and then, the fused features are incorporated into a fully connected layer to produce the ultimate result. Experiments tested on three datasets illustrate the superior results of our MTSFNet compared to other techniques in both subjective and objective indexes. The contributions of our MTSFNet are outlined as follows.

1) A novel HSI CD framework is proposed, which can fully utilize the difference and temporal information from original images. Compared to traditional HCD methods, the proposed MTSFNet can better represent the spectral–spatial–temporal features of input images.
2) A multitemporal symmetric feature extraction module is devised to amalgamate the difference and temporal features, which can greatly boost the separability between changed and unchanged pixels.
3) Extensive experiments conducted on three public database, i.e., Bay Area, Santa Barbara, and Farmland, validate that our MTSFNet achieves outstanding CD performance compared to several recently proposed methods.

The following sections of the article are structured as below. Section II offers a detailed explanation of the designed scheme. Section III analyzes experimental results and ablation experiments. Finally, the conclusion of the article is presented in Section IV.

II. Methodology

Fig. 1 illustrates the schematic of our MTSFNet, which primarily comprises two key components: feature extraction and feature fusion. First, a dual-branch CNN network is used to capture the difference and temporal features. Then, an interactive feature fusion module is devised to integrate the extracted features. Finally, the merged information is

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incorporated into a fully connected layer to produce the ultimate detection result.

A. MTSFNet Framework

The structure of MTSFNet consists of a dual-branch CNN module and a symmetric feature fusion module. Assume $\mathbf{T}_1 \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{T}_2 \in \mathbb{R}^{H\times W\times C}$ be the dual-temporal HSIs, where H and W denote the height and width of the source images, respectively, and C indicates the amount of spectral bands. First, to more effectively capture the changed and temporal information from original images, the subtraction and concatenation operations are performed on the input data $\mathbf{T}_1$ and $\mathbf{T}_2$ so as to obtain changed and temporal images $\tilde{X}_1 \in \mathbb{R}^{H\times W\times C}$ and $\tilde{X}_2 \in \mathbb{R}^{H\times W\times 2C}$, which can be written as follows:

$$\tilde{X}_1 = \mathbf{T}_1 - \mathbf{T}_2 \tag{1}$$

$$\tilde{X}_2 = \mathrm{Cat}(\mathbf{T}_1, \mathbf{T}_2) \tag{2}$$

where $\tilde{X}_1$ represents the changed data, and $\tilde{X}_2$ denotes the temporal data.

Then, the changed and temporal information $\tilde{X}_1$ and $\tilde{X}_2$ are input into a dual-branch three-layer CNNs to extract changed and temporal features

$$\mathbf{X}_1 = F_{\mathrm{CNN}}(\tilde{X}_1) \tag{3}$$

$$\mathbf{X}_2 = F_{\mathrm{CNN}}(\tilde{X}_2) \tag{4}$$

where F_{CNN} denotes the three-layer CNNs operation. $\mathbf{X}_1$ and $\mathbf{X}_2$ represent the extracted features.

Finally, the multitemporal symmetric feature fusion scheme is utilized to amalgamate the extracted features. Additionally, the fused features are incorporated into a fully connected layer to generate the detection map

$$\mathbf{F}_{\mathrm{fused}} = F_f(\mathbf{X}_1, \mathbf{X}_2) \tag{5}$$

where F_f means feature fusion operation, and $\mathbf{F}_{\mathrm{fused}}$ represents the final fused features.

B. Dual-Branch CNN-Based Feature Extraction

For capturing the high-order features from the changed and temporal data, a dual-branch CNN is proposed, which involves convolutional layers, pooling layers, and activation functions. Specifically, given an input data $\tilde{X}_k, k = 1, 2$, the output feature of the lth layer can be calculated using the following equation:

$$\begin{aligned}\mathbf{F}_{i,j}^{l+1,k}(\mathbf{X}_k) &= f\left(\left(\mathbf{W}^{l+1,k} * \mathbf{F}^{l,k}\left(\tilde{X}_{\mathbf{k}}\right)\right)_{i,j} + \mathbf{B}^{l+1,k}\right) \\ &= f\left(\sum_{m=1}^{s_{l+1}}\sum_{n=1}^{s_{l+1}}\sum_{p=1}^{s_{l+1}} \mathbf{W}_{m,n,p}^{l+1,k} \cdot \mathbf{F}_{i+m-1,j+n-1,p}^{l+1,k}\left(\tilde{X}_k\right) + \mathbf{B}^{l+1,k}\right)\end{aligned} \tag{6}$$

where $\mathbf{F}_{i,j}^{l+1,k}(\mathbf{X}_k)$ indicates the feature of the lth layer from the kth branch. $\mathbf{W}^{l+1,k}$ and $\mathbf{B}^{l+1,k}$ stands for the convolution kernel and bias of the kth unit in the $l+1$th layer, respectively.

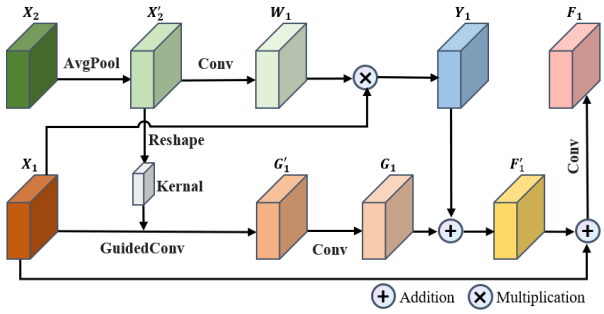


Fig. 2. Architecture of the feature fusion module.

s_{l+1} indicates the dimension of the convolution kernel of layer $l+1$, and $f(\cdot)$ indicates the activation function. For our MTSFNet, the rectified linear unit (ReLU) activation function is utilized, and the input patch size P is set to be 11×11.

C. Symmetric Feature Fusion Module

The symmetric feature fusion module is a crucial component for feature fusion, with the goal of improving the discriminability between changed and unchanged regions. Recent research has demonstrated that the feature representation obtained by CNNs can be enhanced by merging learning mechanisms into the network for capturing spatial dependencies between features [37]. In this work, we seek to incorporate spatial attention to better capture change information by utilizing the changed features $\mathbf{X}_1$ and temporal features $\mathbf{X}_2$. Fig. 2 describes the architecture of the feature fusion module. In more detail, first, $\mathbf{X}_1$ and $\mathbf{X}_2$ are merged together, in which $\mathbf{X}_2$ serves as the guidance layer. To reduce the spatial dimension of $\mathbf{X}_2$, an average pooling operation is applied on $\mathbf{X}_2$

$$\mathbf{X}_2' = \mathrm{AvgPool}(\mathbf{X}_2) \tag{7}$$

where $\mathbf{X}_2'$ denotes the result of average pooling, and AvgPool$(\cdot)$ denotes the average pooling operation.

Then, a convolution operation with 1×1 is performed on $\mathbf{X}2'$ to produce the feature weights. Additionally, an element-wise multiplication operation is employed to combine $\mathbf{W}_1$ and $\mathbf{X}_1$

$$\mathbf{W}_1 = \mathrm{Conv}(\mathbf{x}_2) \tag{8}$$

$$\mathbf{Y}_1 = \mathbf{W}_1 \times \mathbf{X}_1 \tag{9}$$

where $\mathbf{W}_1$ denotes the weight. Meanwhile, the SFFM introduces a kernel generation mechanism to produce kernels for convolution operations, which involves two steps: reshape and linear transformation

$$\mathbf{K} = \mathrm{Reshape}(L(\mathbf{x}_2)) \tag{10}$$

where $\mathbf{K}$ denotes the kernel and L$(\cdot)$ is linear transformation.

Next, the kernel $\mathbf{K}$ is exploited to guide the input feature $\mathbf{X}_1$, yielding the initial feature map $\mathbf{G}_1'$. Moreover, a convolutional operation with a 1×1 kernel is applied on $\mathbf{G}_1'$ to produce the intermediate feature map $\mathbf{G}_1$

$$\mathbf{G}_1' = \mathrm{GuideConv}(\mathbf{X_1}, \mathbf{K}) \tag{11}$$

$$\mathbf{G}_1 = \mathrm{Conv}\left(\mathbf{G}_1', \mathbf{K}\right). \tag{12}$$

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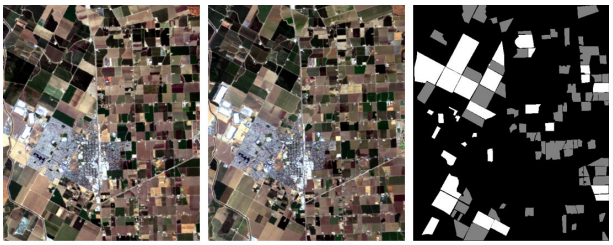


Fig. 3. Bay Area dataset. (a) and (b) Composite image. (c) Reference map.

Finally, the input features $\mathbf{X}_1$ and intermediate features $\mathbf{G}_1$ are integrated together to generate the final features $\mathbf{F}_1$

$$\mathbf{F}_1 = \mathrm{Conv}\left(\mathbf{F}_1' + \mathbf{X}_1\right) \tag{13}$$

where

$$\mathbf{F}_1' = \mathbf{G}_1 + \mathbf{Y}_1. \tag{14}$$

In a similar manner, when the input feature $\mathbf{X}_1$ is used as the guiding image, we can obtain symmetric features denoted as $\mathbf{F}_2$. Moreover, the acquired symmetric features $\mathbf{F}_1$ and $\mathbf{F}_2$ are merged together for the subsequent fully connected layer

$$\mathbf{F}_{\text{fused}} = \mathrm{Conv}(\mathbf{F}_1 + \mathbf{F}_2) \tag{15}$$

where $\mathbf{F}_1$ and $\mathbf{F}_2$ represent the extracted symmetric features.

III. Experiments and Analysis

A. Datasets

In this work, three pairs of datasets are employed to assess the detection effect of our MTSFNet. Each pairs of dataset comprises two HSIs and reference map, in which the HSIs have undergone atmospheric, geometric, and radiometric corrections. The ground truth (GT) uses black to indicate changed pixels and white for unchanged pixels. Unknown pixels are shown in gray. The related information of all adopted datasets are introduced as follows.

The Bayarea dataset is gathered during 2013 and 2015 above Patterson city in California. It encompasses 224 spectral channels and exhibits spatial dimensions of 600 × 500 pixels [38]. Fig. 3 depicts pseudocolor image and GT.

The Santa Barbara dataset comprises two HSIs over Santa Barbara County, captured by the AVIRIS airborne detector in 2013 and 2014. This dataset possesses a spatial resolution of 20 m and spatial size of 984 × 740 pixels. It covers a spectral wavelength range of 0.4 to 2.5 μm with 224 spectral bands [38]. Fig. 4 presents pseudocolor images and GT.

The Farmland dataset is gained by the Earth Observing-1 (EO-1) Hyperion sensor, which is located in a farmland around Yancheng, Jiangsu, China. This dataset is taken on May 2006 and April 2007, respectively. The image comprises 242 spectral channels spanning the range of 0.4–2.5 μm. Spatial resolution is of 30 m. Before experiment, several noise and water absorption channels are eliminated, and the resting 155 spectral bands are preserved. The spatial dimension of each image is of 450 × 140, primarily covering farmland areas [39]. Fig. 5 reports a composite image and a reference map.

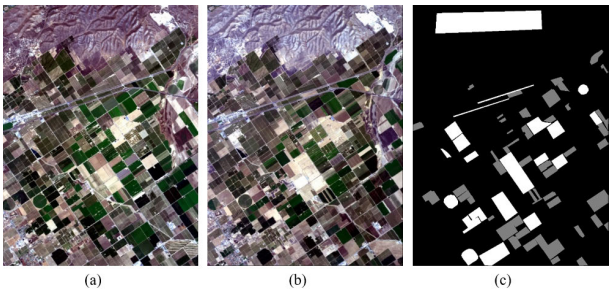


Fig. 4. Barbara dataset. (a) and (b) Composite image. (c) Reference map.

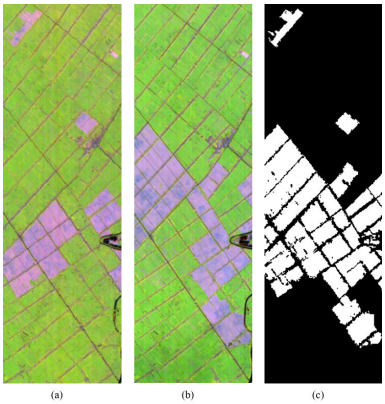


Fig. 5. Farmland dataset. (a) and (b) Composite image. (c) Reference map.

B. Assessment Criteria

To objectively evaluate the effectiveness of various CD methods, several widely used indexes are adopted, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient [40]. The calculation process of all quantitative indices is provided as follows. The OA is computed as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{16}$$

where TP means the quantity of true positive, which refers to the amount of properly detected changed pixels. TN represents the quantity of true negative, which refers to the amount of properly detected unchanged pixels. FP indicates the quantity of false positive, which refers to the amount of mistakenly detected unchanged pixels. FN means the quantity of false negative, which refers to the amount of changed pixels that were mistakenly predicted as unchanged pixels.

The AA is computed as

$$\mathrm{AA} = \frac{\mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) + \mathrm{TN}/(\mathrm{FP} + \mathrm{TN})}{2}. \tag{17}$$

The Kappa coefficient is determined as

$$\mathrm{Kappa} = \frac{\mathrm{OA} - P}{1 - P} \tag{18}$$

$$P = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} + \frac{(\mathrm{TN} + \mathrm{FN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{19}$$

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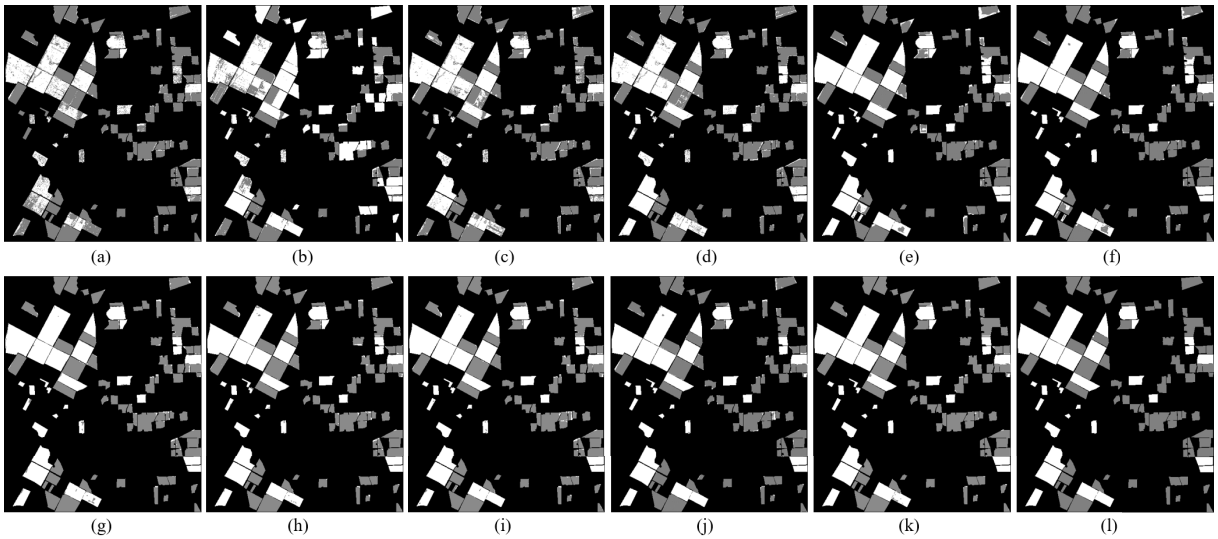


Fig. 6. Comparison of CD maps generated by various methods on the Bay Area dataset. (a) CVA [16]. (b) PCA-KM [41]. (c) IRMAD [15]. (d) SVM [18]. (e) BCNNs [42]. (f) ReCNN [43]. (g) CSANet [44]. (h) MSDFFN [45]. (i) ML-EDAN [46]. (j) TriTF [47]. (k) Proposed. (l) GT.

C. Experimental Settings

In order to measure the efficacy of the proposed MTSFNet, various sophisticated CD models are utilized for comparison, including CVA [16], PCA with k-means clustering (PCA-KM) [41], iteratively reweighted multivariate alteration detection (IR-MAD) [15], SVM [18], bilinear CNNs (BCNNs) [42], recurrent CNN (ReCNN) [43], cross-temporal interaction symmetric attention network (CSANet) [44], multiscale diff-changed feature fusion network (MSDFFN) [45], multilevel encoder–decoder attention network (ML-EDAN) [46], and triplet transformer framework (TriTF) [47]. All experimental results are averaged over ten repetitions to provide a robust assessment of the detection performance across different methods. The detailed description is as below.

1) CVA: CVA is a frequently employed CD technique, which utilizes the Euclidean distance between pairs of pixels to classify them as belonging to changed or unchanged regions.

2) PCA-KM: The PCA-KM algorithm initially applies the PCA approach to decrease the channel dimensionality of the original data. Next, the data with reduced dimensions is clustered with k-means into two groups: changed and unchanged areas.

3) IR-MAD: IR-MAD is an improvement multivariate CD algorithm, where the correlation analysis is used to determine the iterative weight to each pixel.

4) SVM: SVM is a traditional machine-learning method commonly used for ground object classification, which can be directly applied for CD.

5) BCNNs: BCNNs employs two symmetric CNNs to learn feature representations from bitemporal images, fully exploiting the relationship between the two original data through a linear outer product scheme.

6) ReCNN: ReCNN is an end-to-end model that links CNN and RNN. The CNN component generates rich spectral–spatial features, while the RNN component effectively analyzes temporal correlations in dual-temporal images.

7) CSANet: CSANet is an attention-based Siamese network that integrates spatial–spectral–temporal features to achieve CD.

8) MSDFFN: MSDFFN aims to improve feature representation capability through the learning of refined change components across dual-temporal HSIs at various scales.

9) ML-EDAN: ML-EDAN utilizes the hierarchical and temporal features of HSIs to achieve accurate CD. It exploits dual attention mechanisms to capture relevant information and improve CD effect.

The implementation of all the examined approaches is performed on a computer equipped with an NVIDIA GTX 3090 graphics processing unit (GPU). The Adam optimizer [48] is used for model training, with a constant learning rate of 0.001. The training method consists of 150 epochs. In order to provide an equitable comparison, a random selection of 1% of the samples from the Bay Area and Santa Barbara datasets is designated as the training set. Similarly, 10% of the samples from the Farmland dataset are chosen as the training set. The samples at rest are used as the test set.

D. Experimental Results

1) Performance Evaluation on Bay Area and Santa Barbara Datasets: Considering that both the Bay Area and Santa Barbara datasets are acquired by the AVIRIS sensor, their experimental results are together discussed. The two scenes mainly contain urban buildings and farmland. Figs. 6 and 7 depict the experimental results obtained from the Bay Area and Santa Barbara datasets, respectively. The corresponding quantitative evaluations are listed in Table I. The experimental results indicate that unsupervised techniques, including CVA, PCA-KM, and IRMAD, exhibit comparatively lower objective detection performance. This is due to the reason that the spatial information of neighborhood pixels is not leveraged in these techniques. As a result, the resulting detection maps often contain numerous isolated noise points. In comparison, the supervised methods obtain significant improvement.

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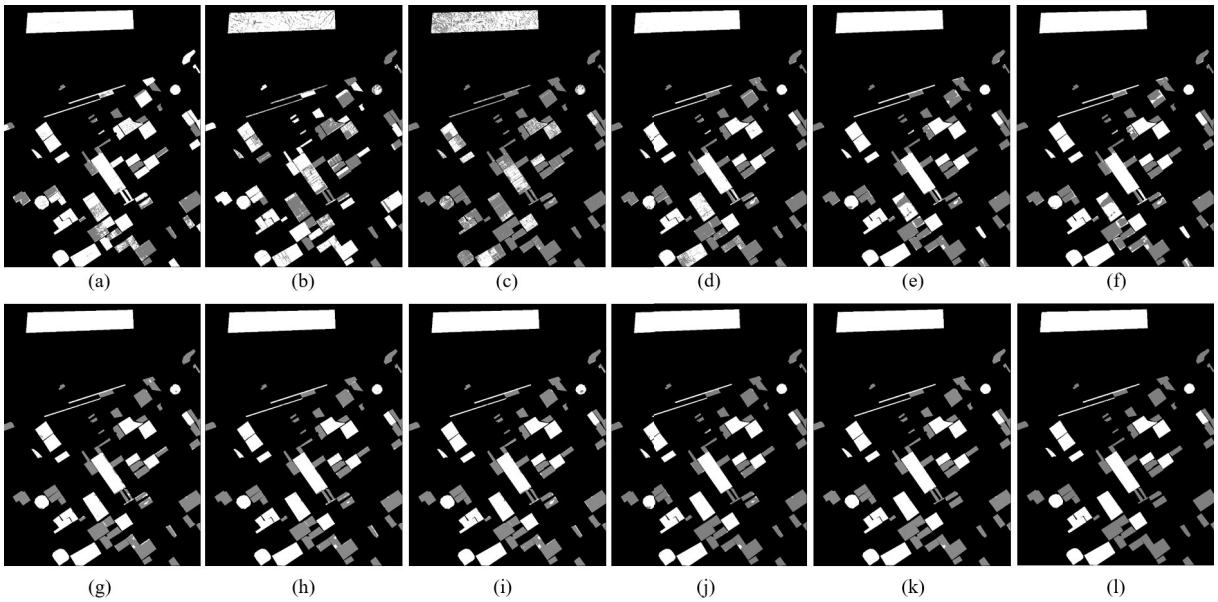


Fig. 7. Comparison of CD maps generated by various methods on the Santa Barbara dataset. (a) CVA [16]. (b) PCA-KM [41]. (c) IRMAD [15]. (d) SVM [18]. (e) BCNNs [42]. (f) ReCNN [43]. (g) CSANet [44]. (h) MSDFFN [45]. (i) ML-EDAN [46]. (j) TriTF [47]. (k) Proposed. (l) GT.

TABLE I
PERFORMANCE COMPARISON OF VARIOUS APPROACHES ON THREE DATASETS

Dataset	Indexes(%)	CVA [16]	PCA-KM [41]	IRMAD [15]	SVM [18]	BCNNs [42]	ReCNN [43]	CSANet [44]	MSDFFN [45]	ML-EDAN [46]	TriTF [47]	Proposed
Bay Area	OA	85.08	68.54	90.43	92.22	95.28	94.41	97.75	98.14	97.96	98.38	**98.93**
	AA	69.55	70.61	83.35	92.21	95.48	94.69	97.74	98.16	98.01	98.42	**98.92**
	Kappa	50.96	38.12	76.27	84.38	90.55	88.83	95.50	96.26	95.92	96.76	**97.80**
Santa Barbara	OA	79.82	56.16	71.45	93.58	95.23	94.09	97.95	98.44	98.27	98.55	**99.52**
	AA	84.63	52.73	77.68	93.11	94.65	93.04	97.56	98.25	98.08	98.34	**99.50**
	Kappa	54.19	57.17	46.54	86.64	89.95	87.42	95.29	96.74	96.37	96.95	**99.00**
Farmland	OA	95.23	87.54	89.32	96.39	96.41	96.13	96.44	97.98	96.69	98.22	**98.40**
	AA	93.81	88.97	86.54	94.27	94.83	95.58	95.72	97.05	95.05	97.64	**98.25**
	Kappa	88.55	66.68	76.60	90.75	91.14	90.17	91.34	95.84	93.41	95.67	**96.13**

The SVM method shows similar results on both datasets, ranking moderately in terms of accuracy among all compared methods. However, the detection performance is low than the deep-learning-based HCD approaches.

For deep-learning-based HCD schemes, they perform well than traditional machine-learning methods on both datasets. The ML-EDAN fully utilizes the hierarchical and temporal features of multitemporal HSIs, which obtains better detection effect over CSANet. The MSDFFN improves the detection results over other CD methods, since it extracts fine-grained change components from dual-temporal HSIs at different scales. By contrast, the proposed method can excels in detecting sporadic and large-area changed pixels. The CD results are more aligned with the GT.

Furthermore, our proposed method demonstrates superior performance on both datasets, as shown in Table I. In comparison to the advanced method, the OA is increased by 0.55% and 0.97% on both datasets, respectively. This further confirms the merit of the proposed MTSFNet with respect to other CD techniques.

2) Performance Evaluation on Farmland Dataset: The visual results produced by all examined schemes on the Farmland dataset are displayed in Fig. 8. The corresponding evaluation metrics acquired by all studied approaches are presented in Table I. The visualization maps obtained by unsupervised methods like CVA, PCA-KM, and IRMAD exhibit serious misclassification. The reason is that these methods cannot well distinguish change and unchanged features without any labeled samples. The CVA performs better than PCA-KM and IRMAD methods, as it has fewer unrecognized changed regions, as depicted in Fig. 8(a). Additionally, the SVM method outperforms the unsupervised methods.

Among all deep-learning methods, BCNNs, ReCNN, CSANet, MSDFFN, ML-EDAN, and TriTF yield similar detection maps. There is a misclassification of unchanging pixels as changed pixels. For objective results, the proposed MTSFNet method attains the highest detection accuracies, including OA, AA, and Kappa, surpassing all other studied techniques. This showcases the superiority of the proposed MTSFNet in both qualitative and quantitative aspects.

E. Ablation Experiments

1) Effectiveness of Feature Fusion Module: To validate the effect of the proposed SFFM, an experiment is run on three datasets. In Fig. 9, the objective metrics of the proposed MTSFNet with or without the SFFM module are presented, where the amount of training samples is same with the

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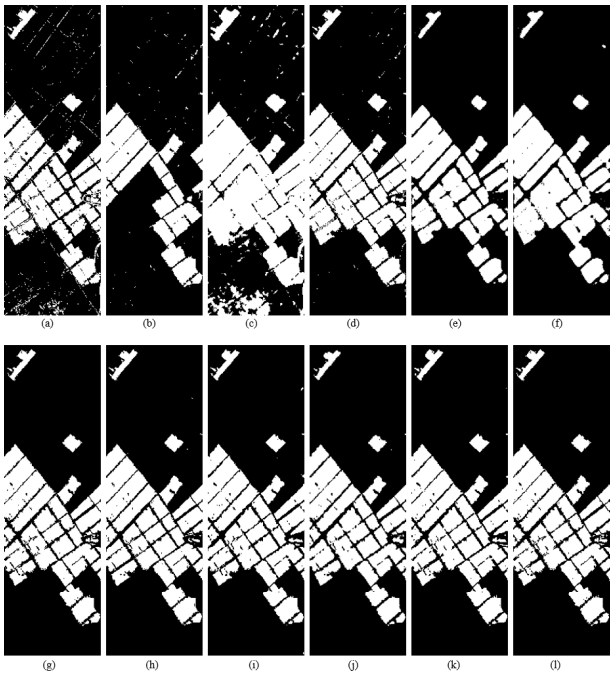


Fig. 8. Comparison of CD maps generated by various methods on the Farmland dataset. (a) CVA [16]. (b) PCA-KM [41]. (c) IRMAD [15]. (d) SVM [18]. (e) BCNNs [42]. (f) ReCNN [43]. (g) CSANet [44]. (h) MSDFFN [45]. (i) ML-EDAN [46]. (j) TriTF [47]. (k) Proposed. (l) GT.

TABLE II
DETECTION PERFORMANCE OF DIFFERENT BRANCHES ON THREE DATASETS

Module			Overall Accuracy (%)		
Difference branch	Temporal branch	SFFM	Bay Area	Barbara	Farmland
✓			98.15	98.27	97.52
	✓		98.02	98.18	97.41
✓	✓	✓	**98.93**	**99.52**	**98.40**

TABLE III
EVALUATION OF COMPUTATIONAL EFFICIENCY ON THREE DATASETS

Methods	Stages	Running time (s)		
		Bay Area	Barbara	Farmland
CVA	Train	-	-	-
	Test	1.42	2.68	1.94
PCA-KM	Train	-	-	-
	Test	1.79	3.29	1.40
IRMAD	Train	-	-	-
	Test	1.58	2.67	0.89
SVM	Train	1.22	1.42	2.07
	Test	38.95	47.14	9.68
BCNNs	Train	2392.87	4348.78	2080.76
	Test	33.82	61.47	29.41
ReCNN	Train	1835.46	3336.21	1596.27
	Test	18.24	33.15	15.86
CSANet	Train	759.94	1430.81	908.29
	Test	56.85	223.03	97.73
MSDFFN	Train	742.15	1422.70	3138.68
	Test	62.07	128.82	13.43
ML-EDAN	Train	1032.88	1356.09	2573.02
	Test	61.90	89.49	25.75
TriTF	Train	13400.64	13157.42	32075.16
	Test	25.47	25.37	23.33
Proposed	Train	147.38	323.33	946.28
	Test	55.23	398.05	10.95

aforementioned experiments. It should be noted that when the SFFM module is not used, a concatenation operation is utilized to merge the multitemporal features. It can be observed from Fig. 9 that the inclusion of the SFFM module obtains an overall improvement in detection effect, which also confirms the effectiveness of the SFFM module.

Furthermore, the discrimination of pixels between changed and unchanged regions is analyzed. Fig. 10 presents t-SNE plots of different features on the Bay Area data. It is evident that the discrimination among pixels in difference features or temporal features is poor. By performing the SFFM, the discrimination of pixels between changed and unchanged regions is greatly improved. This experiment also illustrates that the feature fusion module effectively improves the separability between pixels belonging to unchanged and changed regions, which plays a crucial role in CD.

2) Influence of the Proposed Framework: In this section, the influence of our framework is analyzed. The proposed MTSFNet comprises two CNN branches: the difference branch for extracting differential features and the temporal branch for extracting temporal features. To assess the influence of the two branches, an ablation experiment is examined on the used datasets. As depicted in Table II, the difference branch obtains better detection performance over the temporal branch. In addition, when the two branches are merged, the proposed MTSFNet yields the most favorable detection results across all three datasets. This experiment illustrates that when the complementary information between the difference and temporal branches is taken into consideration, the detection performance can be further optimized.

3) Influence of Varying Numbers of Training Set: In this part, the influence of varying numbers of training set on the CD effect of the proposed MTSFNet is investigated on three datasets. Fig. 11 shows the objective accuracy attained by the proposed MTSFNet with various numbers of training set. For the Bay Area and Santa Barbara datasets, the amount of training set is ranging from 1% to 25%. For the Farmland dataset, the amount of training set is varying from 5% to 25%. As depicted in Fig. 11, As the quantity of training samples increases, the overall accuracies (OAs) of various techniques generally improve. In addition, the OAs achieved by the proposed MTSFNet on all three datasets consistently outperform those of all other compared techniques. This experiment illustrates that the proposed MTSFNet can indeed outperform existing approaches.

4) Influence of Patch Size: The choice of patch size is pivotal in shaping the outcomes of our experiments. To examine its impact on network performance, we tested patch sizes

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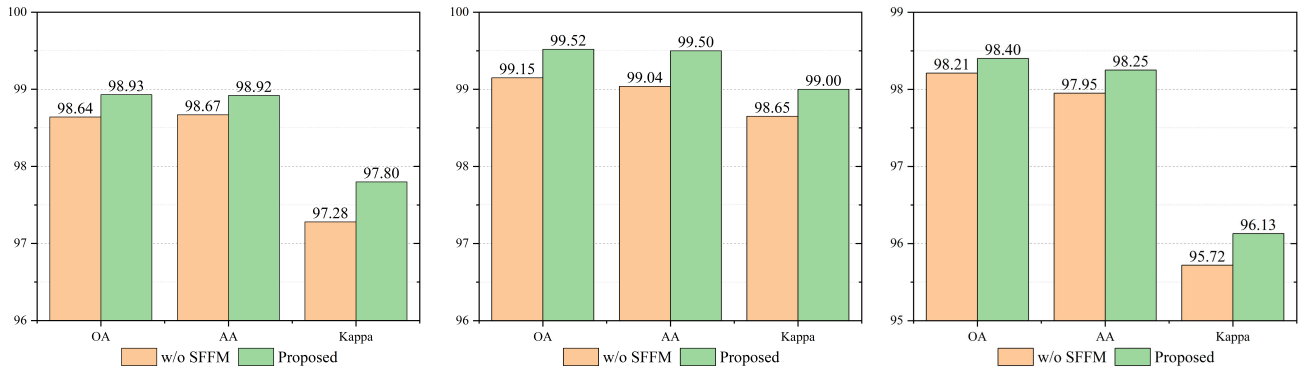


Fig. 9. Ablation results of the MTSFNet on three datasets. (a) Bay Area. (b) Santa Barbara. (c) Farmland.

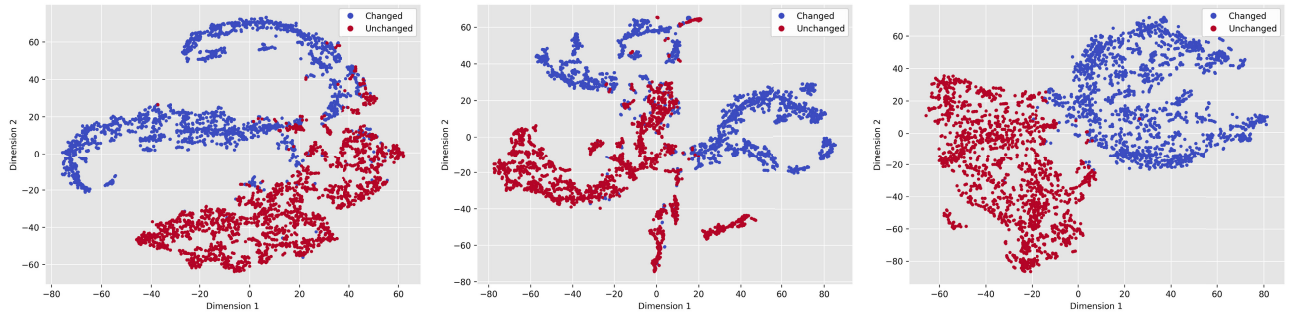


Fig. 10. T-SNE visualization of different features. (a) Difference features. (b) Temporal features. (c) Fused features.

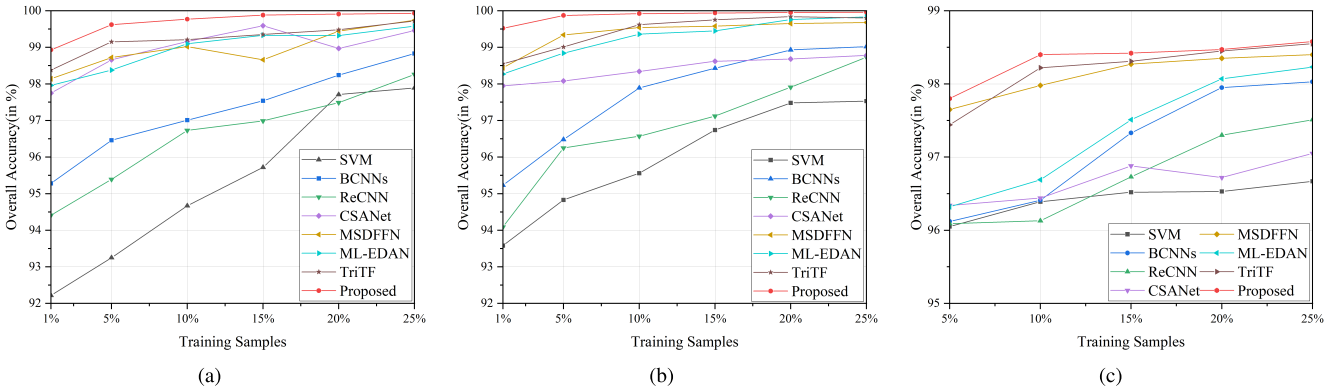


Fig. 11. Effect of the different training ratios on three datasets. (a) Bay Area. (b) Santa Barbara. (c) Farmland.

ranging from 5 × 5 to 13 × 13 across three datasets. Fig. 12 illustrates the OA and Kappa values derived from these experiments. Our analysis reveals that the most effective patch size for maximizing detection accuracy is 11 × 11. This size strikes a balance by providing ample contextual information without introducing redundancy that might compromise performance. Consequently, we have adopted an 11 × 11 patch size configuration in MTSFNet to ensure robust and efficient HCD.

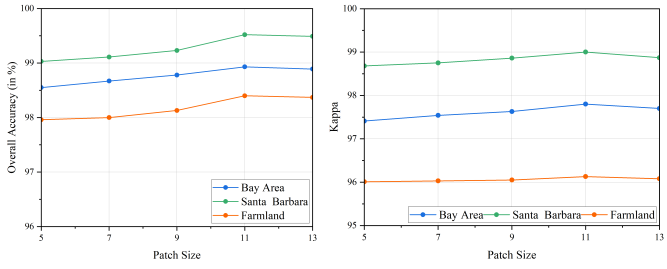


Fig. 12. CD results of MTSFNet with different patch sizes. (a) OA. (b) Kappa.

F. Evaluation of Computational Efficiency

The computational efficiency of various detection methods has been evaluated. Table III summarizes the training and inference times of various detection methods. Traditional machine-learning-based detection approaches, such as CVA, PCA, IR-MAD, and SVM, are computationally efficient yet produce inferior detection outcomes. Conversely, deep-learning-based detection methods typically incur higher computational expenses. Among all deep-learning detection methods, the proposed MTSFNet perform the fastest,

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indicating the competitive advantage in terms of computing burden.

IV. Conclusion

In this work, a novel MTSFNet is proposed for HCD. The proposed approach comprises three main steps. Initially, the difference image and temporal image are extracted from original images. Then, a dual-CNN network is utilized to capture the semantic features. Ultimately, the retrieved features are integrated with the symmetric fusion network and then sent into a fully connected layer to derive the final detection outcomes. The results of comprehensive experiments on three datasets reveal: 1) the proposed method surpasses other advanced HCD methods, including representative machine-learning techniques and recently proposed deep-learning CD techniques; 2) the proposed hyperspectral change framework greatly improves the detection performance over traditional CD frameworks, leveraging both temporal and difference information effectively; and 3) the proposed feature fusion module enhances the differentiation between changed and unchanged pixels.

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